

MATH120-24S2 Discrete Mathematics

Tutorial 5

References

- Topic 7 and 8: Modular arithmetic
- Section 8.4 of Epp

Work on the following problems in groups during the tutorial in week 6, beginning with the marked questions.

1. [B] What is the remainder of 41^{33} when divided by 3?

Solution:

We need to calculate $41^{33} \pmod{3}$. Since $41 \equiv -1 \pmod{3}$, $41^{33} \equiv (-1)^{33} \equiv -1 \pmod{3} \equiv 2 \pmod{3}$. So the remainder is 2.

- ⇒ 2. [M] For each of the following, determine the value(s) of the integer $n > 1$ for which the given congruence is true.

(a) $28 \equiv 6 \pmod{n}$

(b) $68 \equiv 37 \pmod{n}$

Solution:

(a) Note that $28 - 6 = 22$, so n must be a divisor of 22. Since $22 = 2 \cdot 11$, there are $(1 + 1)(1 + 1) = 4$ divisors of 22, namely 1, 2, 11 and 22. However since $n > 1$, then the three possible values for n are 2, 11 and 22.

(b) Note that $68 - 37 = 31$ which is prime. As $n > 1$, then there is only one possible value for n in this case, namely $n = 31$.

- ⇒ 3. Use Fermat's theorem to compute:

(a) [B] $24^{59} \pmod{59}$.

(b) [M] $2^{123} \pmod{29}$.

Solution:

- (a) Note that 59 is prime. By Fermat's little theorem, we have that $24^{58} \equiv 1 \pmod{59}$. Hence, $24^{59} \equiv 24 \pmod{59}$.
- (b) Note that 29 is prime. By Fermat's little theorem, we have that $2^{28} \equiv 1 \pmod{29}$. Hence, we find that

$$2^{123} = 2^{28 \cdot 4 + 11} = (2^{28})^4 \cdot 2^{11} \equiv 1^4 \cdot 2^{11} = 2^{11} \pmod{29}.$$

Now we note that $2^5 = 3 \pmod{29}$, and hence,

$$2^{11} = 2^{5 \cdot 2} \cdot 2 \equiv 3^2 \cdot 2 \pmod{29}.$$

Hence, we find that $2^{123} \equiv 18 \pmod{29}$.

4. [M]

- (a) Prove that if $a, b, c, n \in \mathbb{Z}$ with $a, n > 0$ and $b \equiv c \pmod{n}$, then $ab \equiv ac \pmod{an}$.
- (b) Prove that if $a, b, c, n \in \mathbb{Z}$ with $a, n > 0$ and $b \equiv c \pmod{n}$, then $ab \equiv ac \pmod{n}$.
- (c) Show that the converse of (b) does **not** necessarily hold. That is, find a, b, c, n such that $ab \equiv ac \pmod{n}$ but $b \not\equiv c \pmod{n}$.

Solution:

- (a) Assume that $a, b, c, n \in \mathbb{Z}$ with $a, n > 0$ and $b \equiv c \pmod{n}$. By definition, $b = c + kn$ for some $k \in \mathbb{Z}$. Multiplying both sides of this equation by a , we have

$$\begin{aligned} ab &= a(c + kn) \\ &= ac + k(an). \end{aligned}$$

So, by definition, $ab \equiv ac \pmod{an}$.

- (b) As above, we have that $ab = ac + (ka)n$, so by definition, $ab \equiv ac \pmod{n}$.
- (c) Pick e.g. $a = 2, b = 6, c = 9, n = 6$, then $12 \equiv 18 \pmod{6}$ but $6 \not\equiv 9 \pmod{6}$.

$\implies$ 5. [M]

- (a) Using the Euclidean Algorithm find the inverse of 43 in $\mathbb{Z}_{131}$.
 (b) Use (i) this to solve the equation

$$43x = 6$$

in $\mathbb{Z}_{131}$.

Solution:

- (a) We start by noting that 131 is prime and so $\gcd(43, 131) = 1$, hence 43 is invertible in $\mathbb{Z}_{131}$. However, we still need to work through the Euclidean algorithm to find its inverse:

$$131 = 3(43) + 2$$

$$43 = 21(2) + 1$$

$$2 = 2(1).$$

Reversing the steps above, we get

$$\begin{aligned} 1 &= 43 - 21(2) \\ &= 43 - 21(131 - 3(43)) \\ &= 64(43) - 21(131). \end{aligned}$$

Rearranging the equation above, we get

$$64(43) = 21(131) + 1$$

and hence $64(43) \equiv 1 \pmod{131}$. It follows that $64(43) = 1$ in $\mathbb{Z}_{131}$ and hence $43^{-1} = 64$ in $\mathbb{Z}_{131}$.

- (b) We multiply both sides of the equation with the inverse of 43 and find that

$$64 \cdot 43x = 64 \cdot 6.$$

Now $64 \cdot 6 \equiv 384 \equiv 122 \pmod{131}$. And hence, $x = 122$ in $\mathbb{Z}_{131}$.

6. [M] Show that $8 \mid (3^{2n} - 1)$ for all $n \in \mathbb{Z}^+$,

- (a) using modular arithmetic,
 (b) using induction.

Solution:

- (a) To show that $3^{2n} - 1$ is divisible by 8 (using modular arithmetic), it suffices to show that $3^{2n} - 1 \equiv 0 \pmod{8}$.

We start by noting that

$$\begin{aligned} 3^{2n} &= (3^2)^n \\ &= 9^n. \end{aligned}$$

Since $9 \equiv 1 \pmod{8}$, it follows by Corollary 7.5 that

$$9^n \equiv 1^n \pmod{8}$$

and Theorem 8.1 that

$$\begin{aligned} 9^n - 1 &\equiv 1^n - 1 \pmod{8} \\ &\equiv 1 - 1 \pmod{8} \\ &\equiv 0 \pmod{8}. \end{aligned}$$

Hence $3^{2n} - 1$ is divisible by 8.

- (b) **Proof:** Base Step: For $n = 1$, $3^{2 \times 1} - 1 = 9 - 1 = 8$ and since $8|8$, the base case is true.

Induction step: For some integer $k \geq 1$, assume that the statement is true for $n = k$. That is, $8|(3^{2k} - 1)$. (This is the induction hypothesis.)

We want to prove that the statement is true when $n = k + 1$. That is,

$$8|(3^{2(k+1)} - 1).$$

Now consider $3^{2(k+1)} - 1$:

$$\begin{aligned} 3^{2(k+1)} - 1 &= 3^{2k+2} - 1 \\ &= 3^{2k}3^2 - 1 \\ &= 3^{2k}3^2 - 1 + 3^2 - 3^2 \quad (\text{note that } 3^2 - 3^2 = 0 \text{ so the equation still holds}) \\ &= 3^2(3^{2k} - 1) + 3^2 - 1 \\ &= 9(3^{2k} - 1) + 8 \end{aligned}$$

By the induction hypothesis, $8|(3^{2k} - 1)$ and hence $8|9(3^{2k} - 1)$. Clearly $8|8$. Hence the statement is true when $n = k + 1$.

By the principle of mathematical induction, $8|(3^{2n} - 1)$ for all $n \in \mathbb{Z}^+$.

7. [C] Find an integer x such that $x \equiv 4 \pmod{17}$ and $x \equiv 3 \pmod{13}$.

Solution:

We have to find an integer x such that $x \equiv 4 \pmod{17}$ and $x \equiv 3 \pmod{13}$. We have that $x = 4 + 17k$ for some integer k and $x = 3 + l13$ for some integer l . Subtracting the latter equation from the first yields $-17k + 13l = 1$. So we have to find integer solutions k, l to this equation. Note that $\gcd(13, 17) = 1$, and hence, that we can use the Euclidean algorithm to express 1 as an integer linear combination of 13 and 17.

$$17 = 13 + 4$$

$$13 = 3 \cdot 4 + 1$$

So $1 = 13 - 3 \cdot 4 = 13 - 3 \cdot (17 - 13) = -3 \cdot 17 + 4 \cdot 13$. We find the solution $k = 3$ and $l = 4$ and we see that $4 + 17k = 4 + 51 = 55 = 3 + l13 = 3 + 52$, so $55 \equiv 4 \pmod{17}$ and $55 \equiv 3 \pmod{13}$.

8. (a) [M] Show that if $ab = 0$ in $\mathbb{Z}_{101}$, then $a = 0$ or $b = 0$.
 (b) [C] Use this observation to find all solutions to the equation

$$x^2 + 3x = 99$$

in $\mathbb{Z}_{101}$.

Solution:

- (a) Recall that since 101 is a prime, every non-zero element in $\mathbb{Z}_{101}$ is invertible. Suppose that $ab = 0$ in $\mathbb{Z}_{101}$. If $a = 0$, then we are done, so suppose that $a \neq 0$. This means that a is invertible, so a^{-1} exists. It follows from $ab = 0$ that

$$a^{-1}ab = a^{-1}0.$$

Since $a^{-1}a = 1$ and $a^{-1}0 = 0$ we find that $b = 0$.

- (b) First note that, since $-99 \equiv 2 \pmod{101}$, we can write $x^2 + 3x = 99$ as $x^2 + 3x + 2 = 0$, which easily factors into $(x + 1)(x + 2) = 0$. By (a) we know that this implies that $x + 1 = 0$ or $x + 2 = 0$ in $\mathbb{Z}_{101}$. So we find the solutions $x = 100$ and $x = 99$.

9. [C] What is the last digit of 33^{2021} ?

Solution:

First note that the last digit of a number a is simply the value of $a \pmod{10}$.

Since $33 \pmod{10} \equiv 3 \pmod{10}$ we have that Now

$$\begin{aligned} 33^{2021} \pmod{10} &\equiv 3^{2021} \pmod{10} \\ &\equiv 3 \cdot 3^{2 \cdot 1010} \pmod{10} \\ &\equiv 3 \cdot 9^{1010} \pmod{10} \\ &\equiv 3 \cdot (-1)^{1010} \pmod{10} \\ &\equiv 3 \pmod{10} \end{aligned}$$

So the last digit of 33^{2021} is 3.